

PID Control of a Liquid-Level System

Observing closed-loop feedback control and controller tuning on a GUNT RT 010 level control trainer

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Course	ME 343-004 Mechanical Lab I, NJIT	Instructor	Prof. B. S. Mani · TA: Saba Mirahsani
Team	Team 4: Luis Ramirez, Samuel Cohen, Edwin Ortez, Abhin Ghandi, Ramid Brown	Equipment	GUNT RT 010 liquid-level control trainer, PC control and data-acquisition software
Format	Demonstration-based lab		

PROJECT SUMMARY

This demonstration-based lab introduced closed-loop PID control using a GUNT RT 010 liquid-level control trainer. A pump and valve assembly moved water into and out of a transparent tank, and a controller adjusted the flow to hold the level at a setpoint. The control software showed a live plot of level versus setpoint alongside a system diagram, so the physical tank and the abstract feedback loop could be watched side by side. Setpoint changes and gain adjustments showed how the proportional, integral, and derivative terms each shape the response, and why tuning is essential for accuracy and stability.

1. Setup

The GUNT RT 010 consists of a base unit housing the electronic controller, a pump and valve assembly that moves fluid into and out of the system, and a transparent cylindrical tank with a red float column that rises and falls with the level. Because the tank is transparent, the effect of each controller action could be seen directly rather than only read from a screen.

The trainer was operated from a desktop computer running the control and data-acquisition software, which provided two views: a live graph of liquid level and setpoint over time, and a system diagram showing the flow path with labeled components and live indicators. The TA led the session while the class gathered around a single station.

2. PID Control Principles

PID control is a feedback method that continuously measures the error between a system’s actual value and its setpoint, then computes a corrective output to reduce that error. The controller output is $u(t) = K_p e(t) + K_i \int e(t) dt + K_d de(t)/dt$.

Table 1. Role of each PID term

Term	Responds to	Effect on the response
Proportional (P)	Present error	Fast correction; used alone it leaves a steady-state offset
Integral (I)	Accumulated past error	Eliminates steady-state offset; too much causes overshoot and oscillation
Derivative (D)	Rate of change of error	Anticipates future error and damps overshoot

3. Observations

Setpoint changes. The most instructive part of the session was watching the system respond to a new setpoint. As soon as the target level changed, the controller adjusted the pump and valve to drive the actual level toward it, while the graph showed the current level and the target together as the gap closed.

Tuning. Starting from a well-configured system, the TA changed the controller parameters so the class could see how the response changed and then converged again to a smooth, well-behaved approach to the setpoint. Watching the level settle made the practical meaning of tuning clear: the gains determine whether the system responds quickly, overshoots, or oscillates.

Software interface. The numerical settings for setpoint and controller gains could be changed live, and the system diagram's labeled components and indicators tied the hardware on the bench to the block-diagram representation of a feedback loop from lecture. Seeing the tank, electronics, graph, and schematic work together made it clear how each part contributes to the control loop.

4. Discussion

This lab connected directly to the measurement labs earlier in the semester, which converted temperature, pressure, strain, and other physical quantities into electrical signals, processed them, and displayed them in real time. Here, liquid level played the same role as the measured variable, but the loop was closed: the measurement was fed back to a controller that acted on the system. That shows data acquisition and feedback control are not separate topics but a shared framework used throughout mechanical engineering.

5. Conclusion

The lab provided a practical introduction to closed-loop control. The transparent tank, the guided walkthrough, and the real-time graph made it easy to see how the three PID terms interact and why tuning is crucial for precision and stability. Seeing these concepts in motion made them far easier to understand and remember than working through the mathematics alone.

References

- Holman, J. P. (2012). *Experimental methods for engineers* (8th ed.). McGraw-Hill.
- Zhu, C. (2009). *ME 343 laboratory introduction*. New Jersey Institute of Technology.